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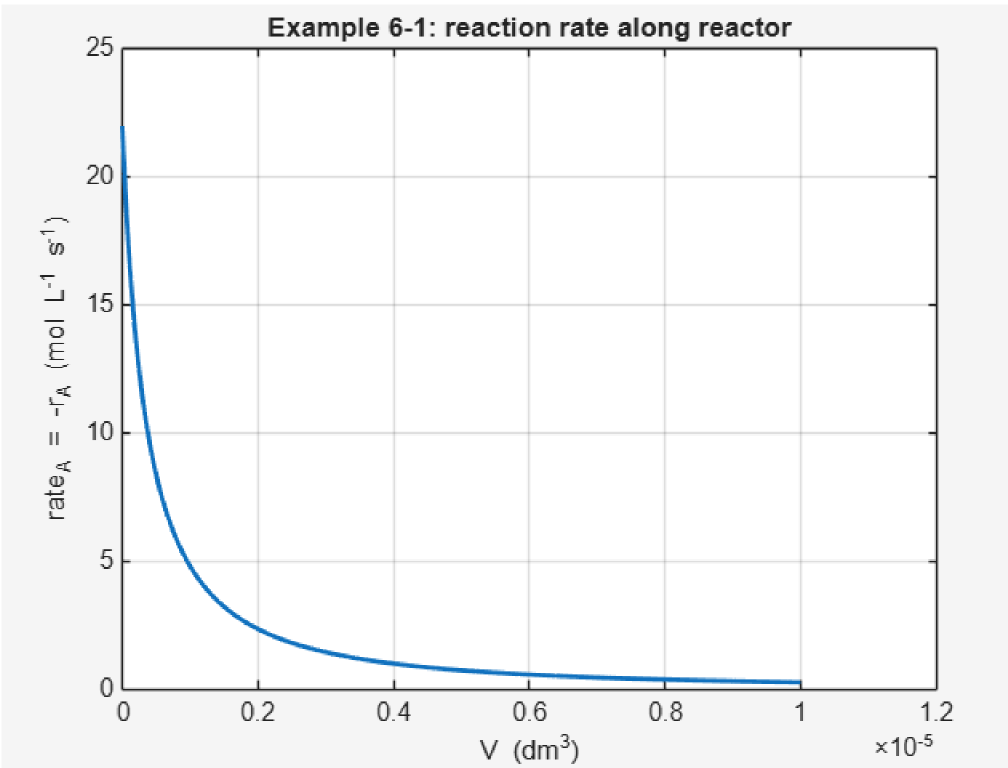
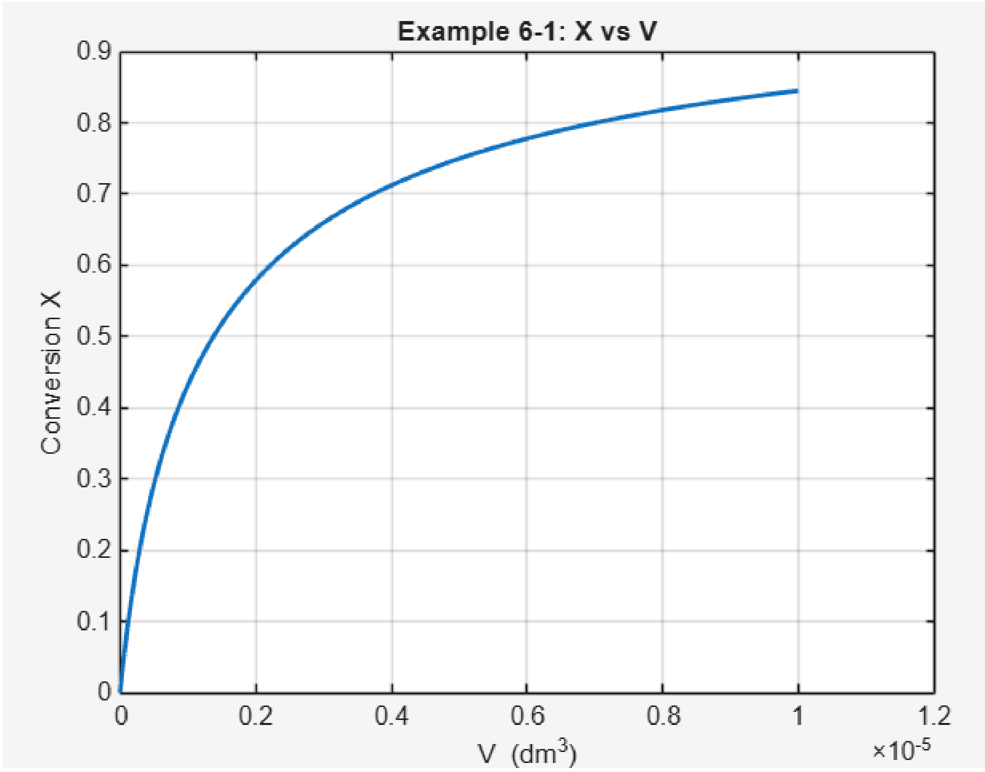
CHE 321 | HW 5

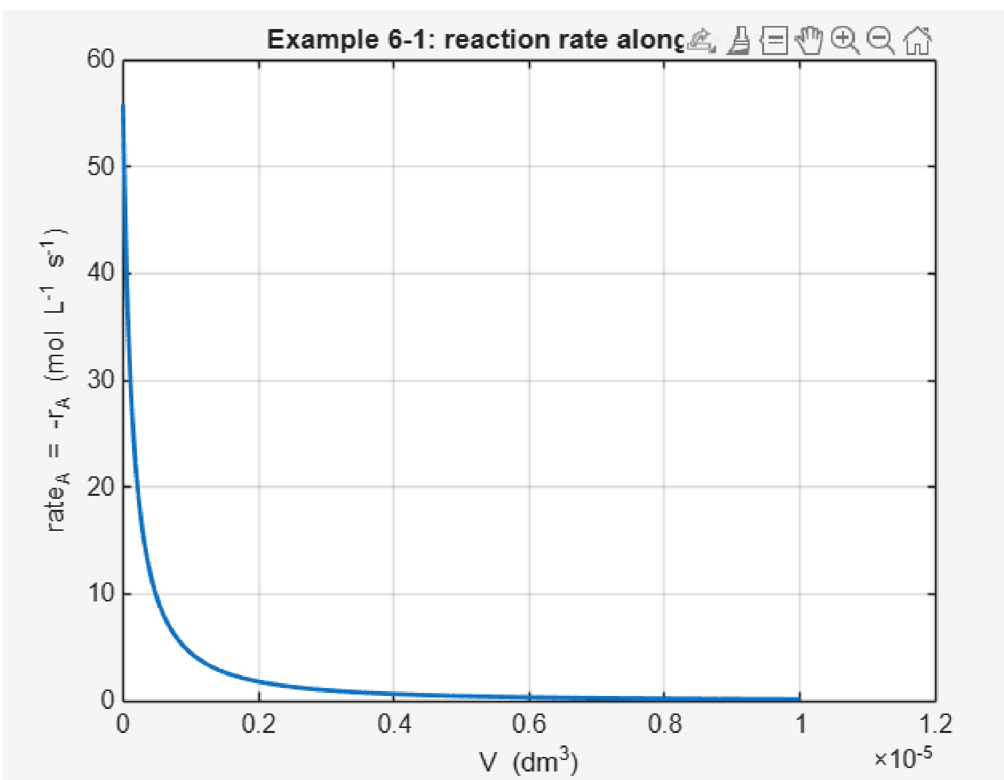
1) A) Example 6-1

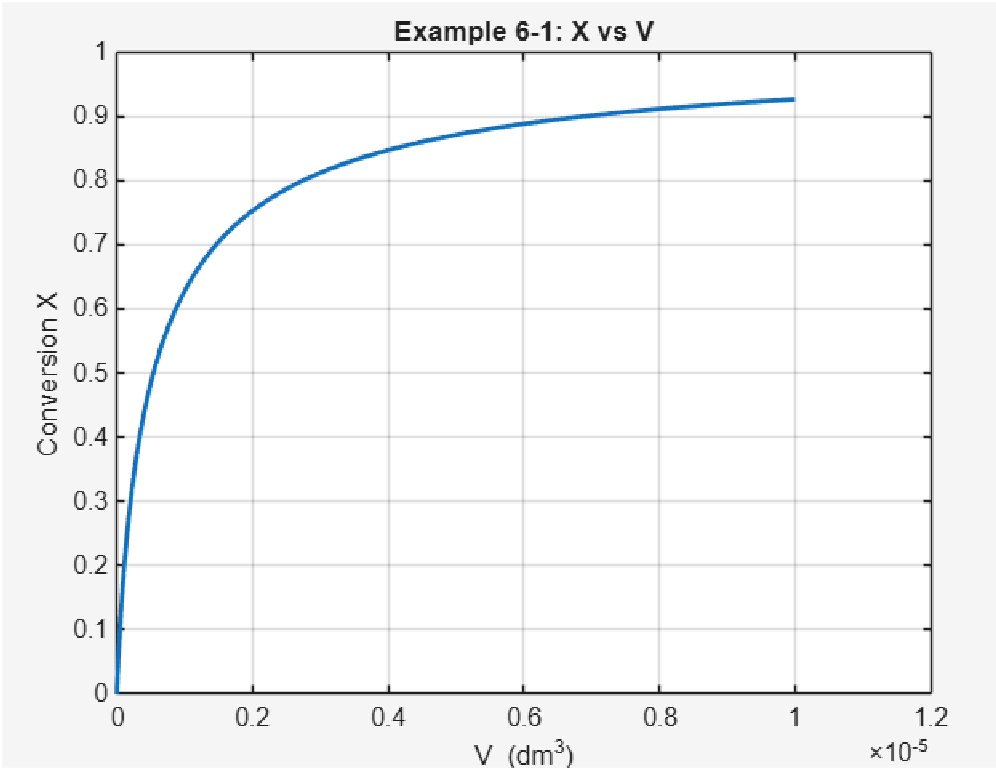
BASELINE GRAPHS



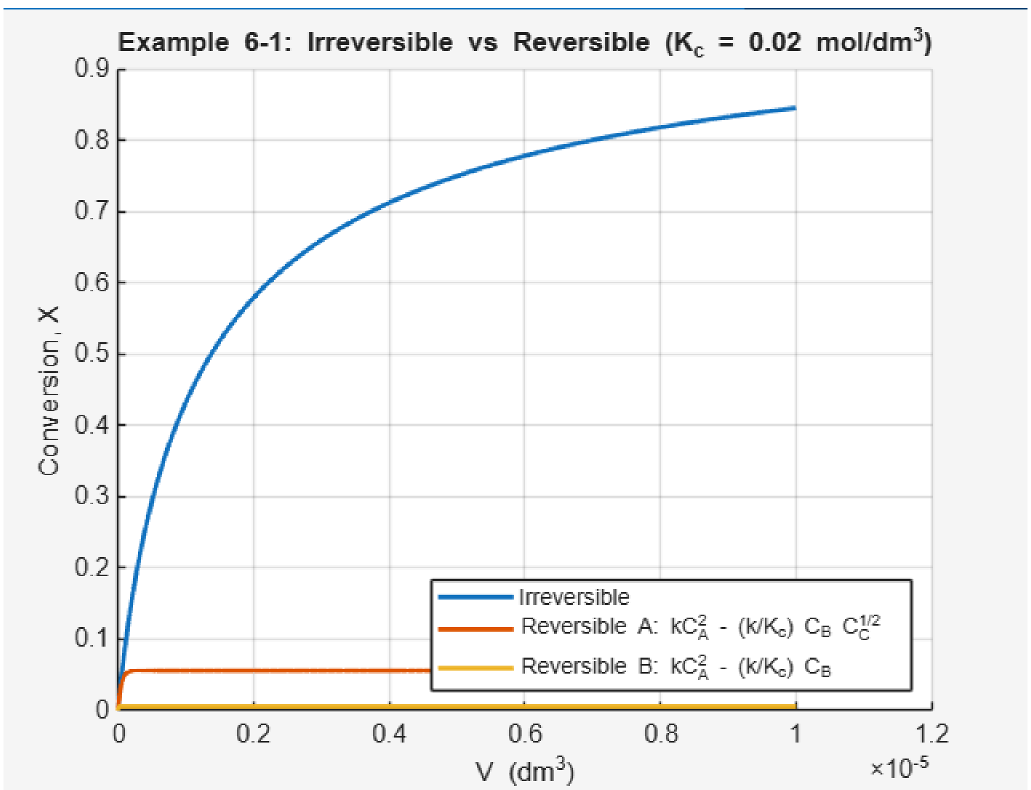


After changes T to 678 and PkPa to 2*1641 (double pressure):



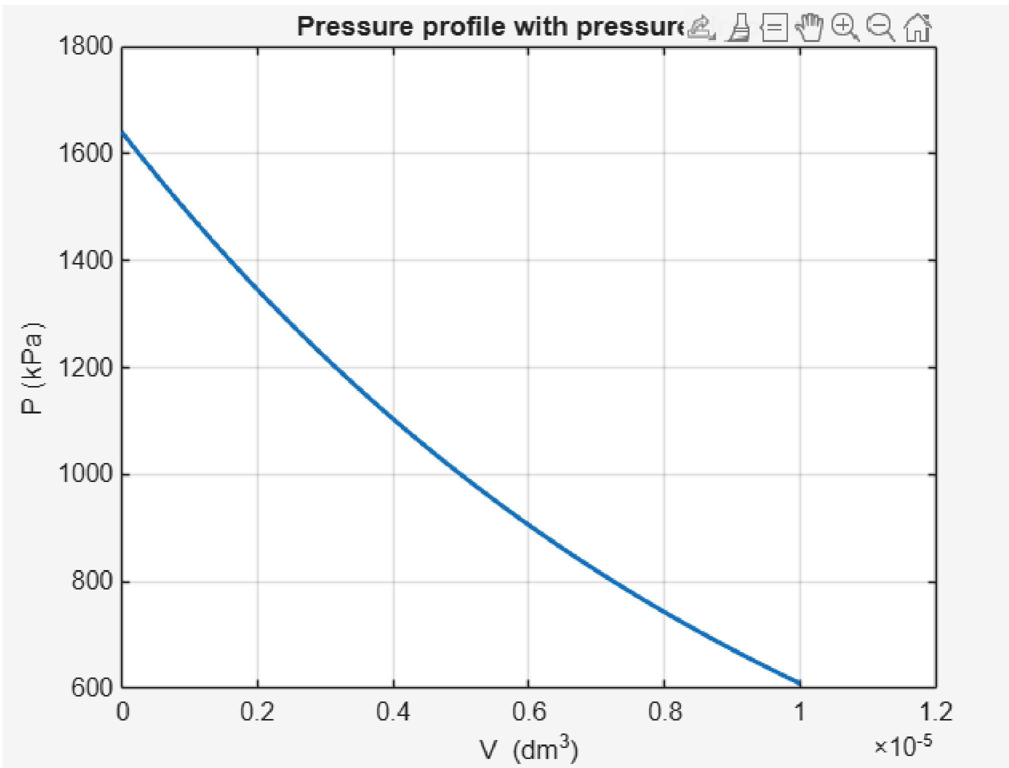


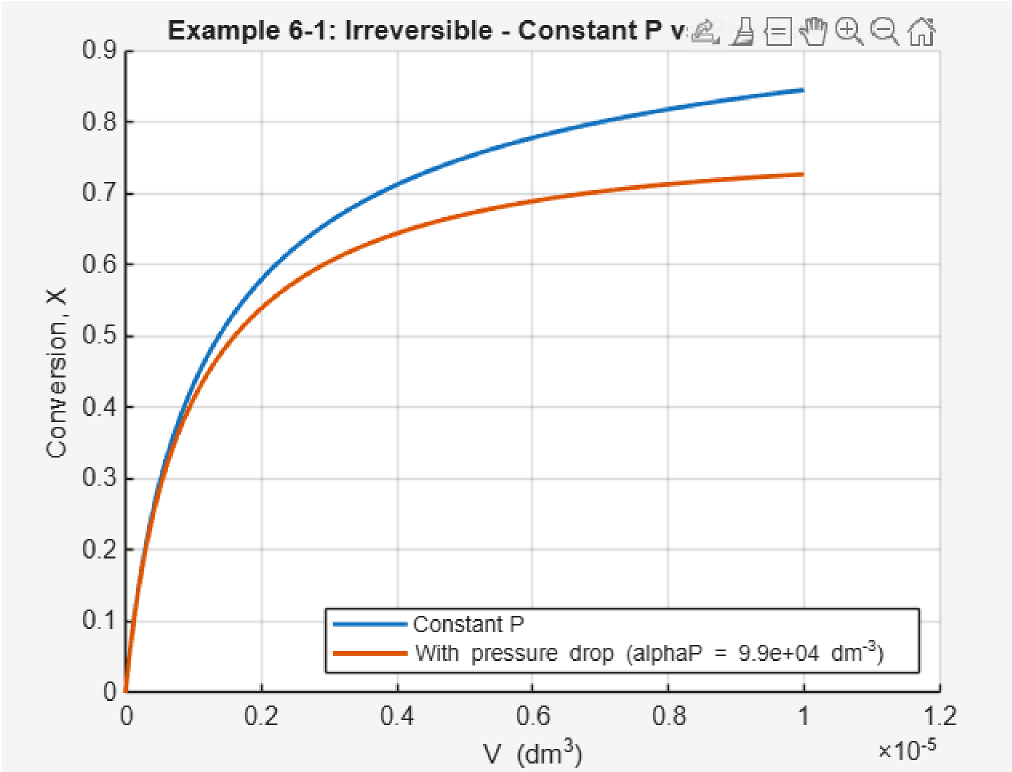
The conversion when the pressure is doubled and temperature lowered is 0.9266.



In figure E6-1.1, F_a decreases steadily while F_b and F_c increase. While in the reversible case, all three curves flatten quickly. F_A levels off, F_B only rises slightly and F_C barely increases. This matches figure E6-1.1 where F_A drops quickly then becomes constant and F_B and F_C plateau.

Reflection: When this reaction is made reversible with K_C=0.02, the conversion approaches a very small equilibrium value very quickly and the molar flow profiles flatten early on in the reactor. This matches figure E6-1.1, which demonstrates that a small equilibrium constant severely limits conversion.





Pressure decreases along the reactor, lowering CA and the second order rate. Conversion drops from roughly 0.84 for the constant pressure case to 0.72 when the pressure drop is implemented. The lower curve confirms that reduced pressure limits the overall reaction progress even though the kinetics are unchanged.

B) Example 6-2

Scanning one at a time, conversion at 500 dm^3 increases with each of k, K_C, C_{T0} and k_C. When k and K_C increase, that's when the largest lift happens due to stronger kinetic drive and a higher equilibrium ceiling. Raising k_C boosts conversion by withdrawing B and driving the reaction forward. With K_C at a maximum, conversion increases with k and k_C, but tapers as the reactor approaches the elevated equilibrium limit. R_B reaches a maximum at a volume where C_A remains appreciable but the buildup of B hasn't suppressed the net rate. Exit conversion rises with decreasing k/k_C and with increasing $(k_T C_{A0})/K_C$ with diminishing returns at the high ratio ends.

Example: for the middle range parameters, what reactor volume does $R_B(V)$ first drop below 50% of its peak value and how does that volume shift when k_C is doubled?

Answer: R_B peaks early where C_A is still high. Doubling k_C removes B faster sothe "50% of its peak" point moves downstream even though the exit F_B may not increase due to stronger stripping. This demonstrates how membrane removal can preserve high net production over a larger portion of the reactor while changing the outlet composition.

C) Example 6-3

Part 1: The maximum allowable feed rate is 2.81×10^3 mol/min. Lowering k slows the formation of B.

Part 2: The maximum feed rate is 2.55×10^4 mol/min. Higher C_{T0} makes the same outlet C_B limit correspond to a smaller X.

Part 3: $F_{Ao,max} = 2.53 \times 10^3$ mol/min. Reflection: reversibility introduces an equilibrium ceiling.

D) In the Wetlands module, increasing rainfall expanded the effective residence time and diluted the solute. This produced lower outlet atrazine concentrations. On the other hand, increasing the evaporation rate reduced the available water and concentrated the contaminant. This caused the outlet concentration to rise. The absolute outlet concentration rose proportionally when the inlet atrazine concentration was increased. However, the fractional removal remained almost constant which is consistent with first-order kinetics. By raising the liquid flow rate, it shortened the residence time and sharply decreased removal efficiency. Overall the simulation showed that pollutant attenuation in wetlands will depend on hydrologic balance and residence time.

E) (1) Nucleation: faster cooling, supersaturation rises quickly. Higher flow rate, the gas spends less time hot and nucleation spikes early but the burst is shorter; Surface growth: faster cooling, limits the time window at elevated temperature. Lower flow, gives particles more time to grow; Flocculation: a strong nucleation burst creates a high number concentration which increases collision frequency. Higher flow, suppresses coagulation, lower flow enhances it.
(i) Helium gives an earlier, higher N(t) peak followed by a faster post-peak decline, argon gives a later, smaller peak with a gentler decline.
(ii) Final diameter is smaller with He and larger with Ar because Ar's slower cooling leaves more time for growth and coagulation.
(iii) The nucleation-rate peak occurs earlier with He than with Ar due to the faster quench.

F) Question: To achieve an 80% conversion of species A in an irreversible second-order liquid phase reaction, a PFR and CSTR are designed. $-r_A=kC^2_A$. Assuming isothermal operation and constant-density flow, if the volumetric flow rate suddenly doubles, which reactor experiences the greater fractional drop in conversion and why?
Why this involves critical thinking: The question requires conceptual reasoning about how reactor type affects sensitivity to flow changes, rather than simply plugging numbers into a formula. We must remember that in a PFR, conversion depends on the integral along the length whereas CSTR depends on the exit-concentration rate law directly.

2) clear; clc; close all;

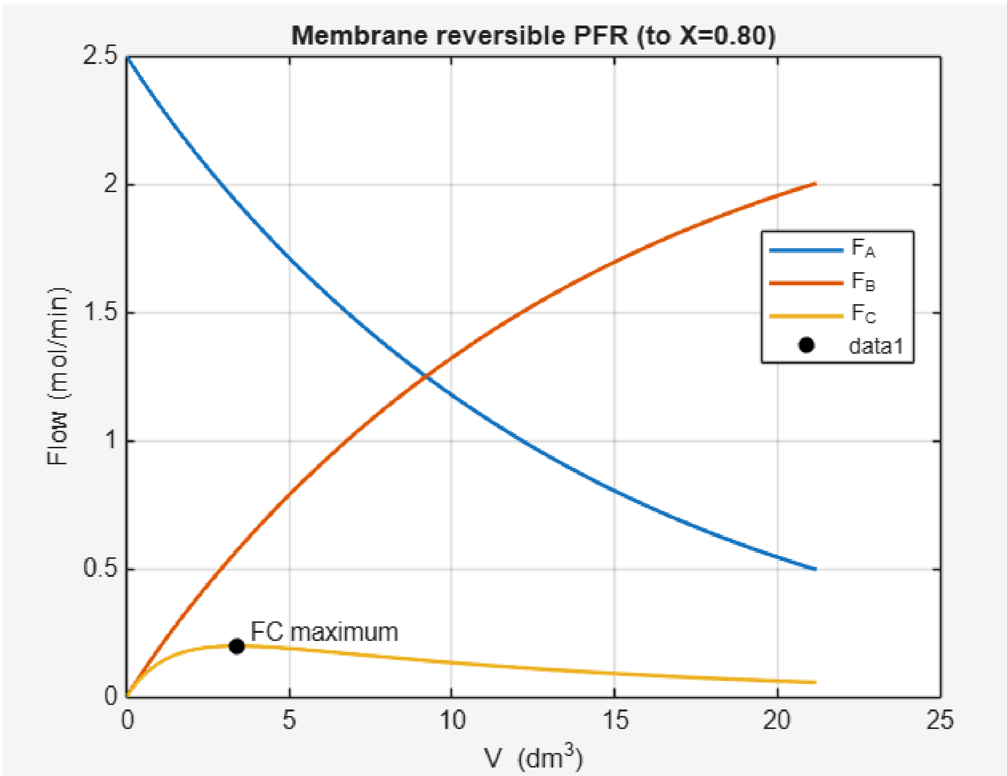
```
% Problem data
T_K   = 300;
P_atm = 10;
R_atm = 0.082057;
CT    = P_atm/(R_atm*T_K);
FA0   = 2.5;
Kc    = 0.025;
kf    = 0.5;
kr    = kf / Kc;

kC_s  = 0.08;
kC    = 60*kC_s;

Xtarget = 0.80;        % integrate until X reaches 0.80

% Helper
fX = @(X) Kc - (CT^2) * (4*X.^3) ./ ((1+2*X).^2 .* (1 - X));
Xeq = fzero(fX, 0.5);
fprintf('Equilibrium conversion Xeq = %.4f\n', Xeq);

% Integration
opts = odeset('RelTol',1e-9,'AbsTol',1e-12);

% Conventional reversible PFR
y0_conv = [FA0; 0; 0];
[V1,Y1] = ode45(@(V,y) rhs_conv(V,y,CT,kf,kr), [0 1e6], y0_conv, opts);
[Vc1, idx1] = stop_at_target(V1,Y1,FA0,Xtarget);
V1 = V1(1:idx1);  Y1 = Y1(1:idx1,:);
FA1 = Y1(:,1); FB1 = Y1(:,2); FC1 = Y1(:,3);
FT1 = FA1 + FB1 + FC1;
X1  = 1 - FA1/FA0;

% Membrane reversible PFR (C removed)
y0_mem = [FA0; 0; 0];
[V2,Y2] = ode45(@(V,y) rhs_mem(V,y,CT,kf,kr,kC), [0 1e6], y0_mem, opts);
[Vc2, idx2] = stop_at_target(V2,Y2,FA0,Xtarget);
V2 = V2(1:idx2);  Y2 = Y2(1:idx2,:);
FA2 = Y2(:,1); FB2 = Y2(:,2); FC2 = Y2(:,3);
FT2 = FA2 + FB2 + FC2;
X2  = 1 - FA2/FA0;

% Find maximum
[FCmax, iFC] = max(FC2);
V_FCmax = V2(iFC);

% Plots
figure('Name','Conversion vs Volume'); hold on; grid on
plot(V1, X1, 'LineWidth', 1.8);
plot(V2, X2, 'LineWidth', 1.8);
yline(Xeq, '--', 'Xeq');
xlabel('V (dm^3)'); ylabel('Conversion, X');
title('P6-5B: X(V) — Conventional vs Membrane (to X=0.80)');
legend('Conventional PFR','Membrane PFR','Location','southeast');
```

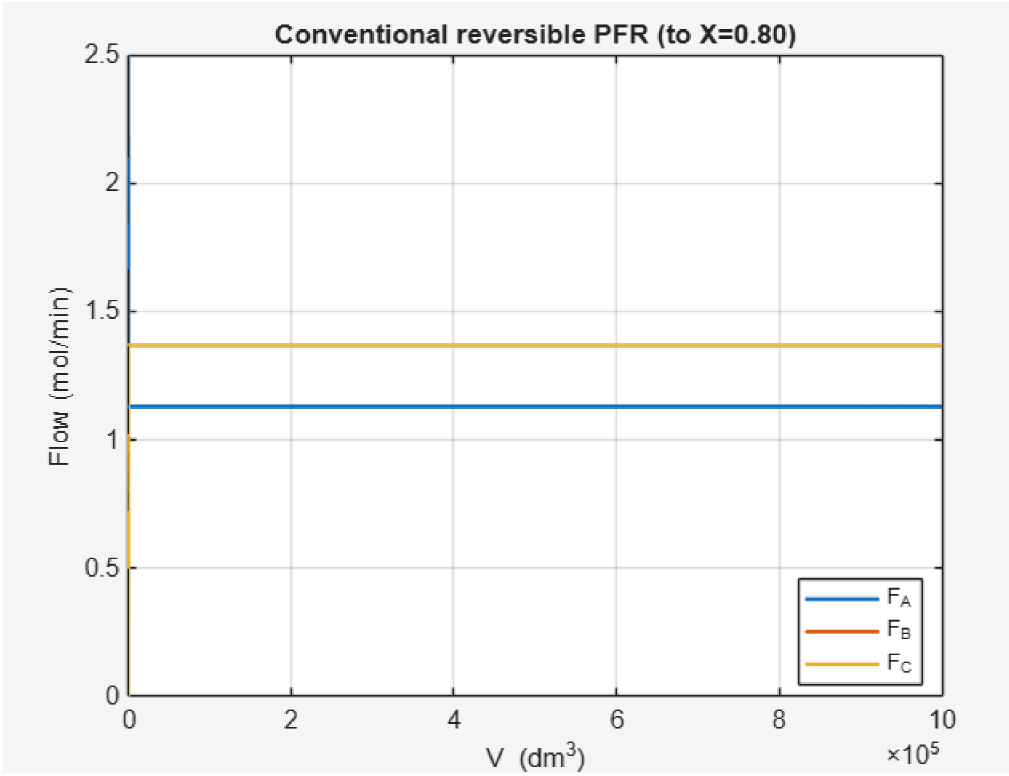
```matlab
figure('Name','Molar flows vs Volume — Conventional');
plot(V1, FA1, V1, FB1, V1, FC1, 'LineWidth', 1.6); grid on
xlabel('V (dm^3)'); ylabel('Flow (mol/min)');
title('Conventional reversible PFR (to X=0.80)');
legend('F_A','F_B','F_C','Location','best');

figure('Name','Molar flows vs Volume — Membrane');
plot(V2, FA2, V2, FB2, V2, FC2, 'LineWidth', 1.6); grid on
xlabel('V (dm^3)'); ylabel('Flow (mol/min)');
title('Membrane reversible PFR (to X=0.80)');
legend('F_A','F_B','F_C','Location','best');
hold on; plot(V_FCmax, FCmax, 'ko', 'MarkerFaceColor','k');
text(V_FCmax, FCmax, '  FC maximum','VerticalAlignment','bottom');

% Solutions
fprintf('\nConventional PFR at X=%.2f: V = %.3g dm^3\n', X1(end), V1(end));
print_summary(V1,FA1,FB1,FC1,FA0,CT,kf,kr,'Conventional');

fprintf('\nMembrane PFR at X=%.2f: V = %.3g dm^3\n', X2(end), V2(end));
print_summary(V2,FA2,FB2,FC2,FA0,CT,kf,kr,'Membrane');
fprintf('  FC shows a maximum of %.4g mol/min at V = %.3g dm^3\n', FCmax, V_FCmax);

% Local functions
function dydV = rhs_conv(~, y, CT, kf, kr)
    FA = y(1); FB = y(2); FC = y(3);
    FT = FA + FB + FC; if FT <= 0, FT = eps; end
    CA = CT*(FA/FT);  CB = CT*(FB/FT);  CC = CT*(FC/FT);
    rA = -( kf*CA - kr*CB*CC^2 );
    dFA = rA;
    dFB = -rA;
    dFC = -rA;
    dydV = [dFA; dFB; dFC];
end

function dydV = rhs_mem(~, y, CT, kf, kr, kC)
    FA = y(1); FB = y(2); FC = y(3);
    FT = FA + FB + FC; if FT <= 0, FT = eps; end
    CA = CT*(FA/FT);  CB = CT*(FB/FT);  CC = CT*(FC/FT);
    rA = -( kf*CA - kr*CB*CC^2 );
    dFA = rA;
    dFB = -rA;
    dFC = -rA - kC*CC;
    dydV = [dFA; dFB; dFC];
end

function [Vcut, idx] = stop_at_target(V,Y,FA0,Xtarget)
    FA = Y(:,1); X = 1 - FA/FA0;
    idx = find(X >= Xtarget, 1, 'first');
    if isempty(idx), idx = numel(V); end
    Vcut = V(idx);
end

function print_summary(V,FA,FB,FC,FA0,CT,kf,kr,label)
    FT = FA+FB+FC; X = 1 - FA/FA0;
```

```
    CA = CT*(FA(end)/FT(end)); CB = CT*(FB(end)/FT(end)); CC = CT*(FC(end)/FT(end));
    rA = -( kf*CA - kr*CB*CC^2 );
    fprintf('[%s]  Vfin=%.3g dm^3,  X_exit=%.4f\n', label, V(end), X(end));
    fprintf('  FA=%.6g  FB=%.6g  FC=%.6g (mol/min)\n', FA(end), FB(end), FC(end));
    fprintf('  CA=%.5f  CB=%.5f  CC=%.5f (mol/dm^3),  -rA_exit=%.4g
(mol/(dm^3·min))\n', CA, CB, CC, -rA);
end
```

A) $X_{eq} = 0.52$ and $X = 0.47$

B)





C) Non-isotheral operation: a modest temperature increase raises the forward rate constant and with product removal, will push the conversion beyond the isothermal single pass limit.